IILM UNIVERSITY Greater Noida

Department of Computer Science and Engineering

Introduction to AI & ML Project

Project Title: YOUTUBE VIDEO

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Abstract

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Abstract

Long-form YouTube videos often contain valuable educational or informational content, but manually watching and converting them into structured notes is time-consuming and inefficient. This project presents an automated **YouTube Video Summarizer and Note Generator** built using a **Large Language Model (LLM)**, specifically **Gemini Pro**.

The system extracts the transcript of a public YouTube video, processes the complete text using an advanced Natural Language Processing (NLP) model, and generates **concise, well-structured study notes or detailed summaries**. A user-friendly **Streamlit interface** allows users to input a YouTube URL, select a preferred language, and view or download the generated notes.

By combining APIs for transcript extraction with LLM-based summarization, this project demonstrates a practical and impactful application of LLMs in improving information accessibility, reducing cognitive load, and saving time for students, educators, and professionals.

Introduction

Background

With the rapid growth of online learning platforms, **YouTube** has become one of the most widely used sources for tutorials, lectures, and educational content. While videos are engaging, they are not always the most efficient format for revision or quick reference. Learners often need **text-based notes** for exam preparation, revision, or documentation.

Recent advancements in **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)**, particularly **Large Language Models (LLMs)** such as Gemini, have enabled machines to understand and generate human-like text. These models can summarize, rephrase, and structure large volumes of text into meaningful and organized content.

This project utilizes these capabilities to convert long YouTube videos into **concise, actionable study notes**, making learning faster and more efficient.

1.2 Problem Statement

Users who rely on YouTube for learning face several challenges:

- Watching long videos requires significant time and attention.
- Important concepts are not always structured or highlighted clearly.
- Manually pausing videos and writing notes is tedious and inefficient.
- There is no built-in feature on YouTube that generates study-ready notes or summaries.

Therefore, there is a need for an **automated system** that can:

- Extract the content of a YouTube video,
- Understand the context,
- Generate structured notes or summaries,
- Support multiple languages for wider accessibility.

1.3 Objectives

The main objective of this project is:

To develop an efficient, automated solution for converting long-form YouTube video content into concise, actionable study notes or detailed summaries using a Large Language Model (LLM).

Specific objectives include:

- To extract raw transcripts from public YouTube videos automatically.
- To use the **Gemini Pro LLM** to summarize and structure the extracted text.
- To present the generated notes through an interactive **Streamlit web interface**.
- To support **multiple languages** for summary generation.
- To demonstrate a real-world, end-to-end application of LLMs in education and content consumption.

1.4 Scope of the Project

The scope of this project includes:

- Summarizing **public** YouTube videos that have transcripts available.
- Generating **text-based summaries or notes** from video transcripts.
- Providing a front-end interface for user interaction.

The project does not include:

- Downloading or processing private or restricted videos.
- Performing complex video/audio processing beyond transcript-level text.
- Real-time streaming summarization.

THEORETICAL BACKGROUND & LITERATURE REVIEW

Overview of Existing Research

The increasing use of YouTube for learning has encouraged research on automatic transcript summarization to help users save time and access key information quickly. Existing studies mainly focus on two strategies: **extractive summarization**, which selects important sentences from the transcript, and **abstractive summarization**, which generates new sentences using natural language understanding.

With the rise of deep learning, **Transformer-based models and LLMs** have shown superior performance due to their ability to capture context and produce coherent human-like summaries. Research also emphasizes the need for transcript preprocessing, as YouTube captions often contain noise, timestamps, and speech errors that affect summary quality. Additionally, practical tools such as web and browser-based summarizers highlight the growing importance of multilingual support and accessibility features for wider usability.

S.no	Citation (short)	Technology used	Dataset (reported)	Error handling / robustness techniques	Results (reported)	Remarks
1	Senthil Pandi et al., 2024 — *Abstractive Summarization of YouTube Videos using LaMini-Flan-T5*	Fine-tuned Transformer LLM (LaMini-Flan-T5), HuggingFace Pipeline, TTS API, Streamlit	YouTube video transcripts retrieved via YouTube API	Preprocessing to remove timestamps/noise, tokenization (NLTK), reduced parameters for low computation	Better summarization than Flan-T5, lower hallucination than Alpaca; efficient with fewer parameters	Focuses on lightweight LLM summarization; adds text-to-speech output + UI
2	Khalil et al., 2025 — *Evaluating the Quality of Medical*	Khalil et al., 2025 — *Evaluating the Quality of Medical*	194 health-related YouTube videos, previously	Transcript cleaning, exclusion of noisy auto-subtitles,	Gemini-1.0 Pro showed almost perfect agreement	Demonstrates LLM capability for expert-level

	Content on YouTube using LLMs	*Content on YouTube using LLMs*	scored by medical experts	token length limits (2000 tokens), standardized temperature, quantization for large models, prompt engineering	(BP Kappa 0.82) with experts; GPT-4o, Mixtral, etc. achieved substantial agreement; prompt engineering improved scoring accuracy	evaluation; highlights need for guided prompts; focuses on *evaluation*, not summarization
3	Singh et al., 2023 — *YouTube Video Summarizer using NLP: A Review*	Review of NLP techniques: TF-IDF, BERT, LSA, SpaCy, deep learning (CNN, LSTM), graph-based summarization, multimodal summarization	Comparative discussion of datasets used in YouTube summarization; includes query-focused datasets, lecture videos, CCTV/YouTube sets	Highlights preprocessing challenges (noise removal, caption cleaning), evaluation metric difficulties, dataset dependency; discusses robustness via multimodal fusion & attention mechanisms	Concludes supervised learning performs better than unsupervised methods when large datasets are available; multimodal summarization improves accuracy	Pure review; does not implement a system. Useful for understanding summarization trends, datasets, and model selection
4	More et al., 2025 — *QuickTube: YouTube Transcript Summarizer Extension*	Browser extension with NLP algorithms; transcript extraction from YouTube API; summarization using	YouTube transcripts (no fixed dataset; real-time extraction from platform)	Transcript preprocessing to remove timestamps & noise; NLP-based key term extraction; user-controlled customization	Provides concise summaries directly in browser; supports multilingual summaries; improves accessibility & user	Focuses on browser extension usability, not deep model optimization; multilingual support makes it

		extractive + abstractive NLP; multilingual output (English, Hindi, Marathi)		of summary length & language	productivity	practical and user-centric
5	Dwany D. S., 2023 — *YouTube Transcript Summarizer*	Seq2Seq (LSTM) + Transformer model; TF-IDF + cosine similarity; Streamlit UI; YouTube Data API V3	News Summary dataset (4515 news articles with headlines, short & full text)	Text cleaning (remove non-alphabetic chars & special symbols), spaCy pipeline, tokenization, train/test split, cosine similarity to choose better model summary	Transformer model outperforms Seq2Seq on similarity metrics; ~15M trainable params used for training	End-to-end web app: generates summary, emails it, plus audio download; strongly focused on accessibility & time-saving use-case

Research Gap

Although several studies have explored automatic text and video summarization, there are still evident gaps when applying these techniques specifically to YouTube educational content. Most existing research focuses on either extractive methods or large deep learning models without prioritizing real-time usability for common users. Many systems rely on limited datasets such as news articles rather than directly training or testing on YouTube transcripts, which significantly differ in structure, noise level, and spoken-language style. Additionally, only a few works address accessibility features, such as providing summaries in multiple languages or offering audio-based outputs for users with disabilities.

Another major gap observed in existing literature is the lack of evaluation techniques that compare summarized outputs generated by different models for the same transcript. While transformer-based approaches show strong results, most studies do not incorporate model comparison mechanisms to ensure the most accurate summary is delivered. Similarly, the integration of user-friendly applications with advanced LLMs remains limited, leaving a gap between research prototypes and practical tools for everyday use.

Therefore, there is a need for a system that combines high-quality LLM summarization, transcript preprocessing, multilingual support, usability features, and reliable model comparison to deliver accurate, accessible, and real-world summaries of YouTube videos.

Conclusion

The review of existing research shows that automatic summarization of YouTube transcripts is an emerging yet evolving field driven by advancements in NLP and deep learning. While current approaches demonstrate the potential of extractive and abstractive models, challenges still remain in handling noisy transcripts, improving summary accuracy, enhancing accessibility, and building practical real-time applications. These gaps highlight the need for a more robust, user-oriented system that effectively combines advanced LLM capabilities with usability features, forming the basis for the proposed YouTube video summarizer in this project.

METHODOLOGY

The proposed system aims to automatically generate concise and structured summaries from YouTube video transcripts using a Large Language Model (LLM). The entire workflow includes transcript extraction, text preprocessing, LLM-based summarization, and the delivery of formatted notes through an interactive web interface. The methodology followed in this system is shown in the sections below.

System Overview

The system takes a YouTube video URL as input and extracts its transcript using an external API. The obtained transcript is then passed to the Gemini Pro LLM model to generate detailed and structured notes. The final output is displayed on a Streamlit web interface where users can preview and download the summary. The summary can also be generated in different languages depending on the user's selection.

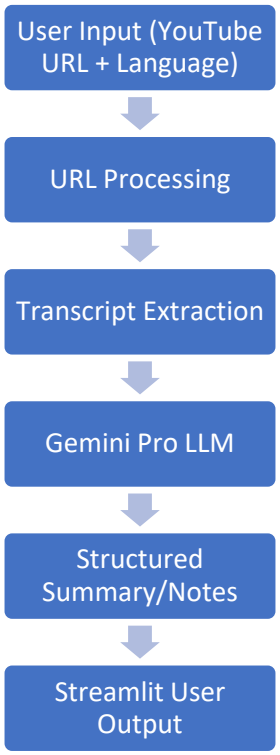
System Architecture

The architecture of the application consists of five key components:

1. **User Interface (Streamlit Web App)**
   Accepts YouTube URL and language selection for summary generation.
2. **URL Processing Module**
   Extracts the video ID from the URL format.
3. **Transcript Extraction Module**
   Uses API services to retrieve the complete transcript of the video.
4. **LLM Summarization Module**
   Processes transcript and generates structured, concise notes using Gemini Pro.
5. **Output Display & Download Module**
   Allows users to view or download summarized notes in text format.

These components collectively ensure end-to-end summarization from input to downloadable output.

Data Flow Diagram



Module Description

Frontend (Streamlit Web App)

- Provides text input for YouTube URL.
- Offers language selection (e.g., English, Hindi, etc.).
- Displays generated summary.
- Allows download of summary file.

Transcript Extraction

- Extracts video ID using string parsing.
- Fetches transcript using third-party APIs.
- Handles exceptions such as:
    - Transcript unavailable
    - Invalid URLs

    - Restricted videos

Transcript Preprocessing

- Removal of timestamps & unnecessary characters (if present).
- Cleaning of ASR noise caused by auto-generated subtitles.
- Optional segmentation for lengthy transcripts.

LLM-Based Summarization (Gemini Pro)

- Uses a pre-trained LLM with high contextual understanding.
- Expert-style prompting to generate note-like summaries.
- Produces structured output including:
    - Headings
    - Bullet points
    - Important terms/definitions

Summary Output & Download

- Summary is displayed in formatted blocks.
- User can export/download as .txt file.

Tools and Technologies Used

Component	**Technology**
Programming Language	Python
Interface	Streamlit
LLM Model	Gemini Pro
Transcript API	Supadata / YouTube Transcript Tool
Environment Management	python-dotenv
Libraries Used	google.genai, streamlit, os

Advantages of the System

- Saves user time while studying lengthy video content.
- Generates structured study notes, not just a paragraph summary.
- Supports multilingual output for better accessibility.
- No manual note-taking required.
- User-friendly interface suitable for students, teachers, and professionals.

Limitations

- No summary can be generated if the transcript is not available.
- Quality of output may vary for poorly captioned videos.
- Depends on API and internet availability.
- Very long transcripts may take more time for processing.

Future Enhancements

- Generate MCQs, flashcards, or key terms from transcript.
- Provide audio narration of the summary.
- Produce topic-wise summaries with timestamps.
- Export summaries to PDF & Docx file formats.
- Real-time summarization with multimodal content (audio + visuals).

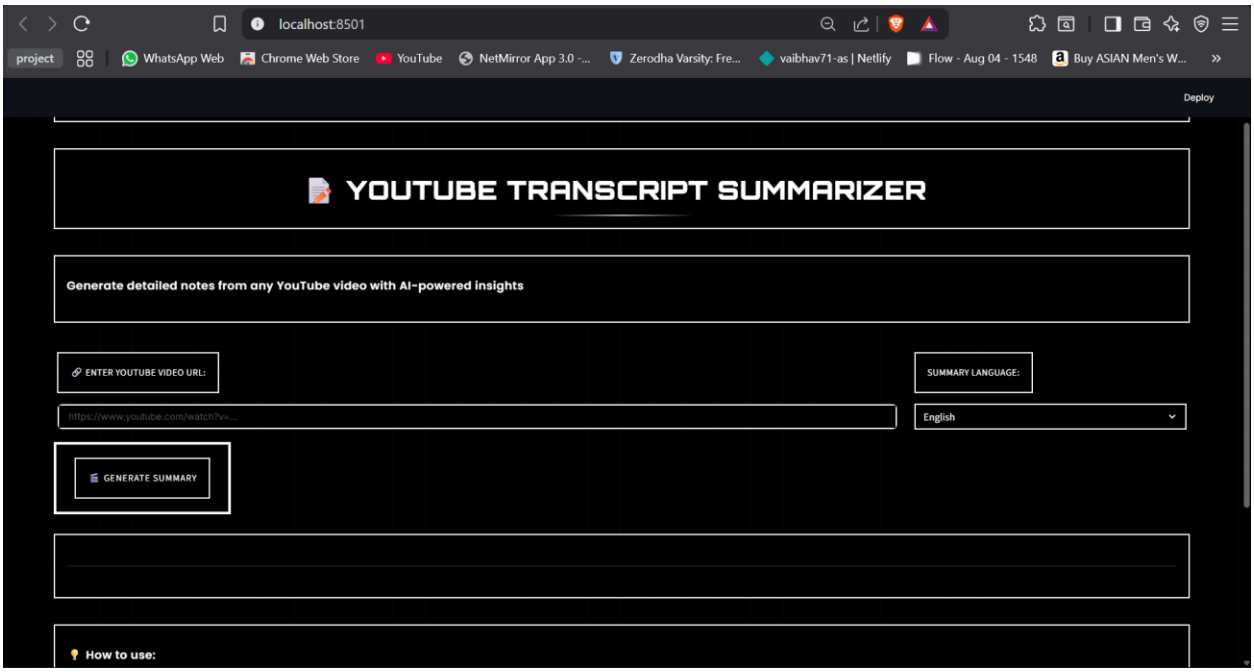
RESULTS & DISCUSSION

The proposed application was executed locally using Streamlit and deployed through a Python environment. Once launched, the system allowed users to provide a YouTube video link, select a preferred language, and generate structured notes using the Gemini Pro LLM. The figures below represent the sequential functioning of the system from deployment to final note generation.

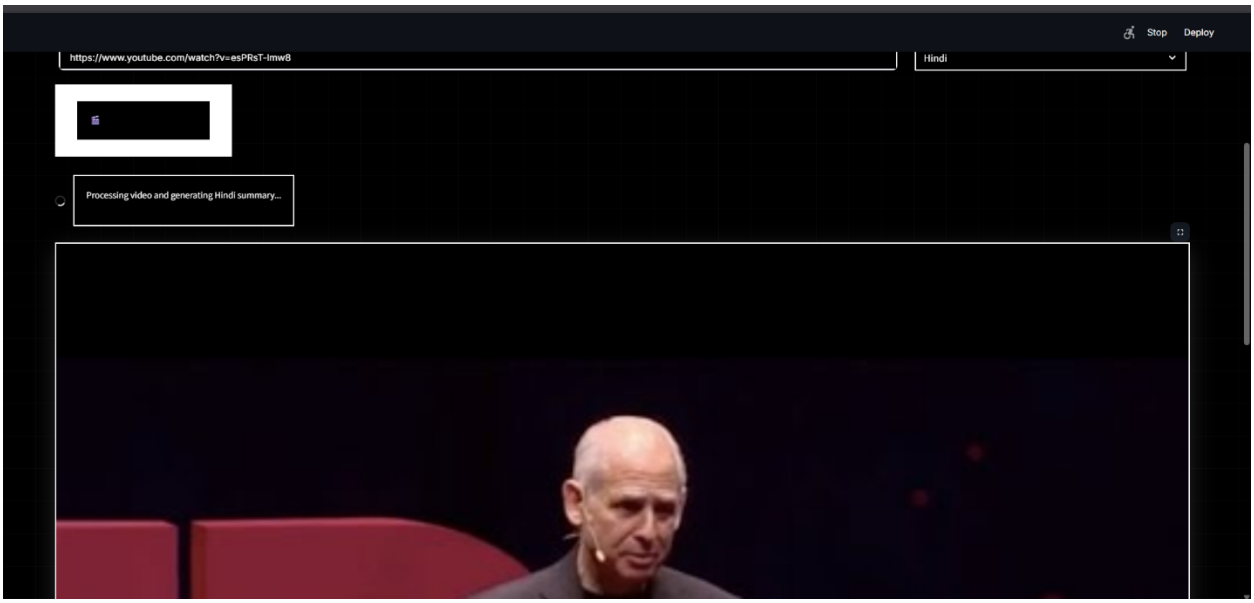
Application Deployment (Terminal / VS Code Execution):

```
21
22  # Custom CSS for black and white minimalist aesthetic - COMPACT VERSION

PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

2025-11-22 19:13:15.363 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-11-22 19:13:15.363 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-11-22 19:13:15.364 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-11-22 19:13:15.364 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
PS C:\Users\Administrator\Documents\git> streamlit run c:\Users\Administrator\Documents\git\project1\gemapp.py

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://192.168.84.116:8501

2025-11-22 19:15:23.708 Please replace `use_container_width` with `width`.
```

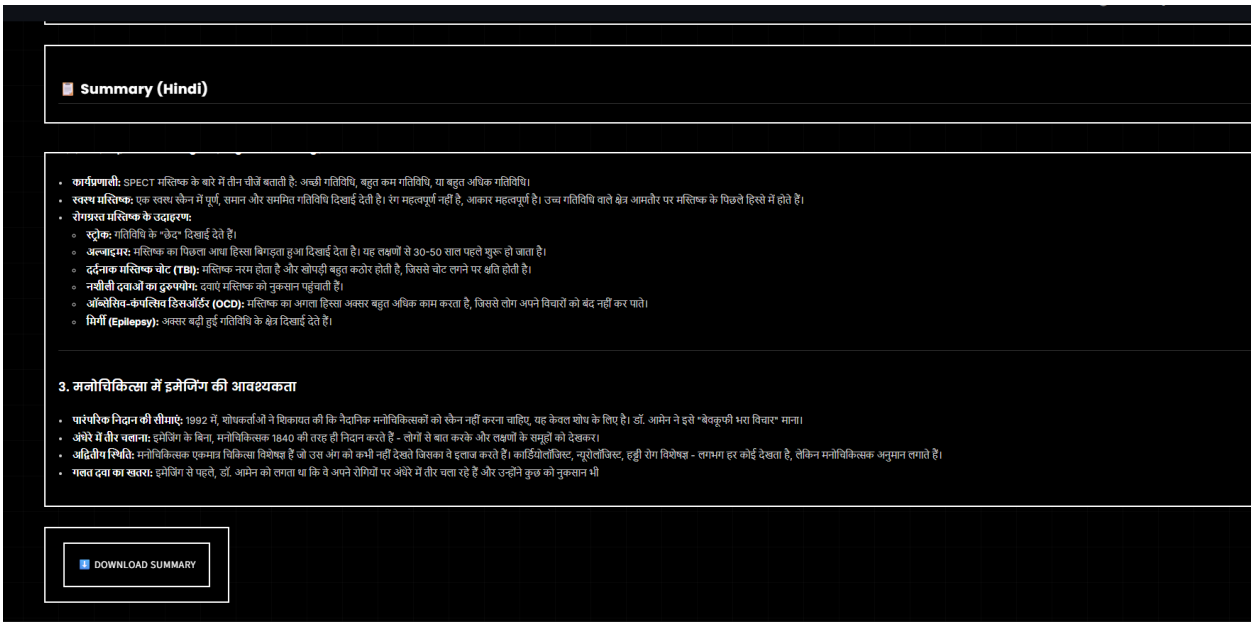
Home Screen Interface:



Processing Stage:



Final Summary Output:



Summary (Hindi)

- **कार्यप्रणाली:** SPECT मस्तिष्क के बारे में तीन चीजें बताती है: अच्छी गतिविधि, बहुत कम गतिविधि, या बहुत अधिक गतिविधि।
- **स्वस्थ मस्तिष्क:** एक स्वस्थ स्कैन में पूर्ण, समान और सममित गतिविधि दिखाई देती है। रंग महत्वपूर्ण नहीं है, आकार महत्वपूर्ण है। उच्च गतिविधि वाले क्षेत्र आमतौर पर मस्तिष्क के पिछले हिस्से में होते हैं।
- **रोगग्रस्त मस्तिष्क के उदाहरण:**
  - **स्ट्रोक:** गतिविधि के "छेद" दिखाई देते हैं।
  - **अल्जाइमर:** मस्तिष्क का पिछला आधा हिस्सा बिगड़ता हुआ दिखाई देता है। यह लक्षणों से 30-50 साल पहले शुरू हो जाता है।
  - **दर्दनाक मस्तिष्क चोट (TBI):** मस्तिष्क नरम होता है और खोपड़ी बहुत कठोर होती है, जिससे चोट लगने पर क्षति होती है।
  - **नशीली दवाओं का दुरुपयोग:** दवाएं मस्तिष्क को नुकसान पहुंचाती हैं।
  - **ऑब्सेसिव-कंपल्सिव डिसऑर्डर (OCD):** मस्तिष्क का अगला हिस्सा अक्सर बहुत अधिक काम करता है, जिससे लोग अपने विचारों को बंद नहीं कर पाते।
  - **मिर्गी (Epilepsy):** अक्सर बढ़ी हुई गतिविधि के क्षेत्र दिखाई देते हैं।

3. मनोचिकित्सा में इमेजिंग की आवश्यकता

- **पारंपरिक निदान की सीमाएं:** 1992 में, शोधकर्ताओं ने शिकायत की कि नैदानिक मनोचिकित्सकों को स्कैन नहीं करना चाहिए, यह केवल शोध के लिए है। डॉ. आमेन ने इसे "बेवकूफी भरा विचार" माना।
- **अंधेरे में तीर चलाना:** इमेजिंग के बिना, मनोचिकित्सक 1840 की तरह ही निदान करते हैं - लोगों से बात करके और लक्षणों के समूहों को देखकर।
- **अद्वितीय स्थिति:** मनोचिकित्सक एकमात्र चिकित्सा विशेषज्ञ हैं जो उस अंग को कभी नहीं देखते जिसका वे इलाज करते हैं। कार्डियोलॉजिस्ट, न्यूरोलॉजिस्ट, हड्डी रोग विशेषज्ञ - लगभग हर कोई देखता है, लेकिन मनोचिकित्सक अनुमान लगाते हैं।
- **गलत दवा का खतरा:** इमेजिंग से पहले, डॉ. आमेन को लगता था कि वे अपने रोगियों पर अंधेरे में तीर चला रहे हैं और उन्होंने कुछ को नुकसान भी

DOWNLOAD SUMMARY

Discussion

The results show that the system is capable of extracting transcripts, processing them through an LLM, and presenting summarized content in a readable and structured manner. The output format closely resembles study notes, making it useful for students preparing from educational videos. Multilingual support further improves accessibility and allows a wide range of users to benefit from the system. The deployment using Streamlit ensures a simple and interactive user experience without the need for complex installations.

CONCLUSION

The YouTube Transcript Summarizer effectively demonstrates the use of Large Language Models for generating concise, accurate, and well-structured notes from long video content. By integrating transcript extraction, multilingual summarization, and a user-friendly Streamlit interface, the system provides a practical solution for students, educators, and professionals who depend on YouTube as a learning source. The generated summaries closely resemble study notes, reducing the need for manual note-taking and improving overall learning efficiency. The results confirm that LLM-based summarization can significantly enhance accessibility, comprehension, and time management in educational content consumption.

5.1 FUTURE SCOPE

- Integration of real-time summarization for YouTube live streams.

- Addition of audio narration for the generated summary to support visually impaired users.
- Generation of MCQs, keywords, and flashcards directly from transcripts.
- Export support for formats such as PDF, Word, and note-taking tools (Notion/OneNote).
- Implementation of multimodal summarization using video + audio + transcript data.
- Mobile application version for quick summarization on smartphones.

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